

correlation function of x x x model

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Contents

1	The XXZ spin-$\frac{1}{2}$ Heisenberg chain	1
2	Quantum inverse scattering problem and correlation functions	4
2.1	Solution of the quantum inverse problem	4
2.2	Action of operators A, B, C, D on a general state	5
2.3	Scalar products	5
2.4	Thermodynamic limit	5
3	Emptiness formation probability	6
4	Correlation function	10

1 The XXZ spin- $\frac{1}{2}$ Heisenberg chain

We give the XXZ Heisenberg $spin - \frac{1}{2}$ finite chain of length M

$$H_{XXZ} = \sum_{m=1}^M \left(\sigma_m^x \sigma_{m+1}^x + \sigma_m^y \sigma_{m+1}^y + \Delta (\sigma_m^z \sigma_{m+1}^z - 1) - \frac{h}{2} \sigma_m^z \right) \quad (1)$$

We can see the ground state is $\begin{cases} \Delta \leq -1 & \text{ferromagnetic} \\ \Delta > -1 & \text{anti-ferromagnetic} \end{cases}$

We focus on the second condition and decompose this domain into a massive regime (for $\Delta > 1$) and a gapless regime (for $-1 < \Delta < 1$) in the thermodynamic limit ($M \rightarrow \infty$)

The R-matrix of the XXZ model is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & b(\lambda, \mu) & c(\lambda, \mu) & 0 \\ 0 & c(\lambda, \mu) & b(\lambda, \mu) & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (2)$$

where

$$b(\lambda, \mu) = \frac{\sinh(\lambda - \mu)}{\sinh(\lambda - \mu + \eta)} \quad c(\lambda, \mu) = \frac{\sinh \eta}{\sinh(\lambda - \mu + \eta)} \quad (3)$$

where

$$\Delta = \frac{1}{2}(q + q^{-1}) \quad q = e^\eta$$

Here the R-matrix act on space $V_1 \otimes V_2$ and depends generically on two spectral parameters λ_1 and λ_2 . We define the *quantum L-operator* acting on space $V_0 \otimes \mathbb{H}_m$

$$L_m(\lambda, \xi_m) = R_{0m}(\lambda - \xi_m) \quad (4)$$

The monodromy matrix is constructed as an ordered product of such L-operators

$$T(\lambda) = R_{0M}(\lambda - \xi_M) \cdots R_{02}(\lambda - \xi_2) R_{01}(\lambda - \xi_1) = \begin{pmatrix} A(\lambda) & B(\lambda) \\ C(\lambda) & D(\lambda) \end{pmatrix} \quad (5)$$

Transfer matrices commute with each other for different values of the spectral parameter λ . They also commute with the Hamiltonian. The Hamiltonian can be reconstructed in terms of the transfer matrix by means of the following "trace identity"

$$H_{XXZ} = 2 \sinh \eta \frac{\partial}{\partial \lambda} \log \mathcal{T}(\lambda)|_{\lambda=\xi_j} + \text{const} \quad (6)$$

Common eigenstates of transfer matrix and Hamiltonian are obtained by $B(\lambda_1) \dots B(\lambda_N) |0\rangle$, where $\{\lambda_k\}$ is a solution of Bethe equations,

$$\frac{a(\lambda_j)}{d(\lambda_j)} \prod_{\substack{k=1 \\ k \neq j}}^N \frac{b(\lambda_j, \lambda_k)}{b(\lambda_k, \lambda_j)} = 1, \quad 1 \leq j \leq N \quad (7)$$

where $a(\lambda) = 1$ and $d(\lambda) = \prod_{i=1}^M b(\lambda, \xi_i)$. The corresponding eigenvalue for the transfer matrix $\mathcal{T}(\mu)$ is then

$$\tau(\mu, \{\lambda_j\}) = a(\mu) \prod_{j=1}^n b^{-1}(\lambda_j, \mu) + d(\mu) \prod_{j=1}^n b^{-1}(\mu, \lambda_j) \quad (8)$$

The logarithmic form of Bethe equations:

$$M p_{0_{tot}}(\lambda_j) + \sum_{k=1}^N \theta(\lambda_j - \lambda_k) = 2\pi n_j, \quad 1 \leq j \leq N, \quad (9)$$

where n_j are integers for N odd and half integer for N even. And

$$p_{0_{tot}}(\lambda) = \frac{i}{M} \ln \frac{d(\lambda)}{a(\lambda)} = \frac{1}{M} \sum_{k=1}^M p_0(\lambda - \xi_k + \eta/2) \quad (10)$$

$$p_0(\lambda) = i \ln \frac{\sinh(\lambda - \frac{\eta}{2})}{\sinh(\lambda + \frac{\eta}{2})} \quad (11)$$

$$\theta(\lambda) = i \ln \frac{\sinh(\eta + \lambda)}{\sinh(\eta - \lambda)} \quad (12)$$

In the thermodynamic limit ($M \rightarrow \infty$), we give the Lieb equation for the ground state:

$$\rho_{tot}(\alpha) + \int_{-\Lambda}^{\Lambda} K(\alpha - \beta) \rho_{tot}(\beta) d\beta = \frac{p'_{0_{tot}}(\alpha)}{2\pi} \quad (13)$$

where

$$\begin{aligned} \alpha &= \lambda & \text{for } -1 < \Delta < 1 \\ \alpha &= i\lambda & \text{for } \Delta > 1 \end{aligned} \quad (14)$$

And

$$\begin{aligned} K(\alpha) &= \frac{\partial}{\partial \alpha} (-\theta(\lambda(\alpha)) / 2\pi) = \frac{\sin 2\zeta}{2\pi \sinh(\alpha + i\zeta) \sinh(\alpha - i\zeta)} \\ p_0(\alpha) &= \frac{\partial}{\partial \alpha} p_{0_{tot}}(\lambda(\alpha)) = \frac{\sin \zeta}{\sinh\left(\alpha + i\frac{\zeta}{2}\right) \sinh(\alpha - i\zeta/2)} \end{aligned} \quad (15)$$

for $-1 < \Delta < 1$, with $\zeta = i\eta$ And

$$\begin{aligned} K(\alpha) &= \frac{\partial}{\partial \alpha} (-\theta(\lambda(\alpha)) / 2\pi) = \frac{\sinh 2\zeta}{2\pi \sin(\alpha + i\zeta) \sin(\alpha - i\zeta)} \\ p'_0(\alpha) &= \frac{\partial}{\partial \alpha} p_{0_{tot}}(\lambda(\alpha)) = \frac{\sinh \zeta}{\sin\left(\alpha + i\frac{\zeta}{2}\right) \sin\left(\alpha - i\frac{\zeta}{2}\right)} \end{aligned} \quad (16)$$

for $\Delta > 1$, with $\zeta = -\eta$, with

$$p'_{0_{tot}}(\alpha) = \frac{1}{M} \sum_{i=1}^M p'_0(\alpha - \beta_k - i\frac{\zeta}{2}) \quad (17)$$

where $\beta_k = \xi_k$ in the domain $-1 < \Delta < 1$, and $\beta_k = i\xi_k$ in the domain $\Delta > 1$. The integration limit Λ is equal to $\frac{\pi}{2}$ and to ∞ for $-1 < \Delta < 1$. In the homogeneous model where all parameter ξ_k are equal to $\frac{\eta}{2}$, the density for the ground state is given by

$$\rho(\alpha) = \frac{1}{2\zeta \cosh\left(\frac{\pi\alpha}{\zeta}\right)} \quad \text{for } -1 < \Delta < 1 \quad (18)$$

$$\rho(\alpha) = \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \frac{e^{2in\alpha}}{\cosh(n\zeta)} \quad \text{for } \Delta > 1 \quad (19)$$

And the solution of inhomogeneous model is given by

$$\rho_{tot}(\alpha) = \frac{1}{M} \sum_{i=1}^M \rho\left(\alpha - \beta_k - i\frac{\zeta}{2}\right) \quad (20)$$

where the inhomogeneous parameter are contained in the region $-\zeta < \text{Im}\beta_j < 0$.

2 Quantum inverse scattering problem and correlation functions

We want to compute the general matrix element of products of local spin operators between two Bethe states $\langle 0 | \prod_j C(\mu_j)$ and $\prod_k B(\lambda_k) | 0 \rangle$.

$$\langle 0 | \prod_j C(\mu_j) \left(\prod_i \sigma_i^{\epsilon_i} \right) \prod_k B(\lambda_k) | 0 \rangle \quad (21)$$

where $\sigma_i^{\epsilon_i}, \epsilon_i \in \{+, -, z\}$ are pauli spin operators at site i .

2.1 Solution of the quantum inverse problem

Theorem 2.1. *Local spin operators at a given site i of the inhomogeneous XXX or XXZ Heisenberg chain are given by:*

$$\sigma_i^- = \prod_{\alpha=1}^{i-1} (A + D)(\xi_\alpha) \cdot B(\xi_i) \cdot \prod_{\alpha=i+1}^N (A + D)(\xi_\alpha) \quad (22)$$

$$\sigma_i^+ = \prod_{\alpha=1}^{i-1} (A + D)(\xi_\alpha) \cdot C(\xi_i) \cdot \prod_{\alpha=i+1}^N (A + D)(\xi_\alpha) \quad (23)$$

$$\sigma_i^z = \prod_{\alpha=1}^{i-1} (A + D)(\xi_\alpha) \cdot (A - D)(\xi_i) \cdot \prod_{\alpha=i+1}^N (A + D)(\xi_\alpha) \quad (24)$$

Proof is by taking x_0 with $\sigma_0^-, \sigma_0^+, \sigma_0^z$ respectively for the Lemma below

Here we give the k-point spin-spin correlation function between two Bethe states. Indeed, for any integer k and any subset $\{i_j\}_{1 \leq j \leq k}$ of $\{1 \dots N\}$, with the convention $i_1 \leq i_2 \leq i_3 \dots \leq i_k$, the correlation function for spins at sites $i_1, \dots, i_k$ between two Bethe states $\langle 0 | C(\mu_1) \dots C(\mu_{n_1})$ and $B(\lambda_1) \dots B(\lambda_{n_2}) | 0 \rangle$ can be written into the following form:

$$\begin{aligned} & \langle 0 | C(\mu_1) \dots C(\mu_{n_1}) \sigma_{i_1}^{\epsilon_1} \sigma_{i_2}^{\epsilon_2} \dots \sigma_{i_k}^{\epsilon_k} B(\lambda_1) \dots B(\lambda_{n_2}) | 0 \rangle \\ &= \prod_{\alpha=1}^{i_1-1} \prod_{j=1}^{n_1} b^{-1}(\mu_j, \xi_\alpha) \cdot \prod_{\alpha=i_k+1}^N \prod_{j=1}^{n_2} b^{-1}(\lambda_j, \xi_\alpha) \\ & \times \langle 0 | C(\mu_1) \dots C(\mu_{n_1}) \cdot X^{\epsilon_1}(\xi_{i_1}) \cdot \prod_{\alpha=i_1+1}^{i_2-1} (A + D)(\xi_\alpha) \cdot X^{\epsilon_2}(\xi_{i_2}) \dots \\ & \times \dots \prod_{\alpha=i_k+1}^{i_k-1} (A + D)(\xi_\alpha) \cdot X^{\epsilon_k}(\xi_{i_k}) \cdot B(\lambda_1) \dots B(\lambda_{n_2}) | 0 \rangle \end{aligned} \quad (25)$$

where $\epsilon_i, 1 \leq j \leq k$ takes the values $+, -,$ or z , X^{ϵ_j} being respectively to $C, B, A - D$.

2.2 Action of operators A, B, C, D on a general state

$$\langle 0 | \prod_{k=1}^N C(\lambda_k) A(\lambda_{N+1}) = \sum_{a'=1}^{N+1} a(\lambda_{a'}) \frac{\prod_{k=1}^N \sinh(\lambda_k - \lambda_{a'} + \eta)}{\prod_{\substack{k=1 \\ k \neq a'}}^{N+1} \sinh(\lambda_k - \lambda_{a'})} \langle 0 | \prod_{\substack{k=1 \\ k \neq a'}}^{N+1} C(\lambda_k) \quad (26)$$

$$\langle 0 | \prod_{k=1}^N C(\lambda_k) D(\lambda_{N+1}) = \sum_{a=1}^{N+1} d(\lambda_a) \frac{\prod_{k=1}^N \sinh(\lambda_a - \lambda_k + \eta)}{\prod_{\substack{k=1 \\ k \neq a}}^{N+1} \sinh(\lambda_a - \lambda_k)} \langle 0 | \prod_{\substack{k=1 \\ k \neq a}}^{N+1} C(\lambda_k) \quad (27)$$

2.3 Scalar products

Theorem 2.2. *Let $\{\lambda_1, \dots, \lambda_n\}$ be a solution of Bethe equations*

$$\frac{d(\lambda_j)}{a(\lambda_j)} \prod_{k \neq j} \frac{b(\lambda_k, \lambda_j)}{b(\lambda_j, \lambda_k)} = 1, \quad 1 \leq j \leq n$$

and $\{\mu_1, \dots, \mu_n\}$ be an arbitrary set of parameters. Then the scalar product

$$S_n(\{\mu_j\}, \{\lambda_k\}) = \langle 0 | \prod_{j=1}^n C(\mu_j) \prod_{k=1}^n B(\lambda_k) | 0 \rangle \quad (28)$$

can be represented as a ratio of two determinants

$$S_n(\{\mu_j\}, \{\lambda_k\}) = S_n(\{\lambda_k\}, \{\mu_j\}) = \frac{\det \mathbf{T}(\{\mu_j\}, \{\lambda_k\})}{\det \mathbf{V}(\{\mu_j\}, \{\lambda_k\})} \quad (29)$$

where T and V are $n \times n$ matrices defined as

$$T_{ab} = \frac{\partial}{\partial \lambda_a} \tau(\mu_b, \{\lambda_k\}), \quad V_{ab} = \frac{1}{\phi(\mu_b - \lambda_a)}, \quad 1 \leq a, b \leq n \quad (30)$$

When the two states are equal, one obtain the Gaudin formula for the norm of a Bethe state

$$\langle 0 | \prod_{j=1}^N C(\lambda_j) \prod_{k=1}^N B(\lambda_k) | 0 \rangle = \sinh^N \eta \prod_{\alpha \neq \beta} \frac{\sinh(\lambda_\alpha - \lambda_\beta + \eta)}{\sinh(\lambda_\alpha - \lambda_\beta)} \det \Phi'(\{\lambda_\alpha\}) \quad (31)$$

where Φ' is a $N \times N$ matrix whose elements are given by

$$\Phi'_{ab} = -\frac{\partial}{\partial \lambda_b} \ln \left(\frac{a(\lambda_a)}{d(\lambda_a)} \prod_{\substack{k=1 \\ k \neq a}}^N \frac{b(\lambda_a, \lambda_k)}{b(\lambda_k, \lambda_a)} \right) \quad (32)$$

2.4 Thermodynamic limit

We have the property

$$\frac{1}{M} \sum_{j=1}^N f(\alpha_j) = \int_{-\Lambda}^{\Lambda} f(\alpha) \rho_{tot}(\alpha) d\alpha + O(M^{-\infty}) \quad (33)$$

The Gaudin matrix in the thermodynamic limit

$$\Phi'_{ab}(\alpha) = -2i\pi M \left(\delta_{ab} \rho_{tot}(\alpha_a) + \frac{1}{M} K(\alpha_a - \alpha_b) \right) + O(M^{-\infty}) \quad (34)$$

$$for -1 < \Delta < 1 \quad (35)$$

$$\Phi'_{ab}(\alpha) = 2\pi M \left(\delta_{ab} \rho_{tot}(\alpha_a) + \frac{1}{M} K(\alpha_a - \alpha_b) \right) + O(M^{-\infty}) \quad (36)$$

$$for \Delta > 1 \quad (37)$$

3 Emptiness formation probability

The emptiness formation probability is the probability to detect a ferromagnetic domain of length m in the antiferromagnetic ground state:

$$\tau(m) = \frac{\langle \psi_g | \prod_{j=1}^m \frac{1}{2}(1 - \sigma_j^z) | \psi_g \rangle}{\langle \psi_g | \psi_g \rangle} \quad (38)$$

We have for the operator $\frac{1}{2}(1 - \sigma_j^z)$ by the quantum inverse scattering problem

$$\frac{1}{2}(1 - \sigma_j^z) = \prod_{k=1}^{j-1} (A(\xi_k) + D(\xi_k)) D(\xi_j) \prod_{k=j+1}^M (A(\xi_k) + D(\xi_k)) \quad (39)$$

So

$$\tau(m) = \phi(\{\lambda\}) \frac{\langle 0 | \prod_{a=1}^N C(\lambda_a) \prod_{j=1}^m D(\xi_j) \prod_{a=1}^N B(\lambda_a) | 0 \rangle}{\langle 0 | \prod_{a=1}^N C(\lambda_a) \prod_{a=1}^N B(\lambda_a) | 0 \rangle} \quad (40)$$

where $\phi_m(\{\lambda\})$ is eigenvalue of tranfer matrix

$$\phi_m(\{\lambda\}) = \prod_{j=1}^m \prod_{a=1}^N \frac{\sinh(\lambda_a - \xi_j)}{\sinh(\lambda_a - \xi_j + \eta)} \quad (41)$$

Using relation (27) we get

$$\begin{aligned} \langle 0 | \prod_{k=1}^N C(\lambda_k) \prod_{j=1}^m D(\lambda_{N+j}) &= \sum_{a_1=1}^{N+1} \sum_{\substack{a_2=1 \\ a_2 \neq a_1}}^{N+2} \dots \\ &\times \sum_{\substack{a_m=1 \\ a_m \neq a_1 \dots a_{m-1}}}^{N+m} G_{a_1 \dots a_m}(\lambda_1 \dots \lambda_{N+m}) \\ &\times \langle 0 | \prod_{\substack{k=1 \\ k \neq a_1, \dots, a_m}}^{N+m} C(\lambda_k) \end{aligned} \quad (42)$$

where the function G can be written as

$$G_{a_1, \dots, a_m}(\lambda_1, \dots, \lambda_{N+m}) = \prod_{j=1}^m d(\lambda_{a_j}) \frac{\prod_{b=1}^{N+j-1} \sinh(\lambda_{a_j} - \lambda_b + \eta)}{\prod_{\substack{b=1 \\ b \neq a_1, \dots, a_j}}^{N+j} \sinh(\lambda_{a_j} - \lambda_b)} \quad (43)$$

To proceed our calculation, we have to calculate the following "normalized" product:

$$\mathcal{S}(\{\lambda_1, \dots, \lambda_{N-m}, \xi_1, \dots, \xi_m\}, \{\lambda\}) = \frac{\langle 0 | \prod_{b=1}^{N-m} C(\lambda_b) \prod_{k=1}^m C(\xi_k) \prod_{k=1}^N B(\lambda_k) | 0 \rangle}{\langle 0 | \prod_{k=1}^N C(\lambda_k) \prod_{k=1}^N B(\lambda_k) | 0 \rangle} \quad (44)$$

By relation (29) and Gaudin formulas (31), we obtain:

$$\begin{aligned} & \mathcal{S}(\{\lambda_1, \dots, \lambda_{N-m}, \xi_1, \dots, \xi_m\}, \{\lambda\}) \\ &= \prod_{j,k=1}^m \frac{\sinh(\lambda_{N-m+k} - \lambda_{N-m+j})}{\sinh(\xi_k - \xi_j)} \prod_{j=1}^m \prod_{k=1}^{N-m} \frac{\sinh(\lambda_k - \lambda_{N-m+j})}{\sinh(\lambda_k - \xi_j)} \\ & \times \prod_{a=1}^N \prod_{k=1}^m \frac{\sinh(\lambda_a - \xi_k + \eta)}{\sinh(\lambda_a - \lambda_{N-m+k} + \eta)} \frac{\det \Psi'(\{\lambda\}, \{\xi\})}{\det \Phi'(\{\lambda\})} \end{aligned} \quad (45)$$

where Φ' is the Gaudin matrix. And for Ψ' we have

$$\begin{aligned} \Psi'_{ab} &= \Phi'_{ab} & b \leq N-m \\ \Psi'_{ab} &= \frac{\sinh \eta}{\sinh(\lambda_a - \xi_{b+m-N}) \sinh(\lambda_a - \xi_{b+m-N} + \eta)} & b > N-m \end{aligned} \quad (46)$$

As The Gaudin matrix is invertible, we get for the determinants

$$\frac{\det \Psi'(\{\lambda\}, \{\xi\})}{\det \Phi'(\{\lambda\})} = \det \left(\Phi'^{-1}(\{\lambda\}) \Psi'(\{\lambda\}, \{\xi\}) \right) \quad (47)$$

The first $N-m$ columns of the matrix $\Phi'^{-1} \Psi'$ are the unit matrix columns:

$$(\Phi'^{-1} \Psi')_{ab} = \delta_{ab}, \quad b \leq N-m \quad (48)$$

Using the representation for the Gaudin matrix in the thermodynamic limit 2.4 and the Lieb equation (13), we obtain:

$$\frac{1}{M} \sum_{b=1}^N \Phi'_{ab} \frac{\tilde{\rho}(\lambda_b - \xi_k + \frac{\eta}{2})}{\tilde{\rho}_{tot}(\lambda_b)} = \frac{\sinh \eta}{\sinh(\lambda_a - \xi_k) \sinh(\lambda_a - \xi_k + \eta)} + O(M^{-\infty}) \quad (49)$$

where

$$\begin{cases} \tilde{\rho}(\lambda) = \rho(\lambda), & -1 < \Delta \leq 1, \\ \tilde{\rho}(\lambda) = i\rho(i\lambda), & \Delta > 1. \end{cases} \quad (50)$$

Thus we obtain

$$(\Phi'^{-1} \Psi')_{ab} = \frac{\tilde{\rho}(\lambda_a - \xi_{b+m-N} + \frac{\eta}{2})}{M \tilde{\rho}_{tot}(\lambda_a)} + O(M^{-\infty}), \quad b > N-m \quad (51)$$

Finally the determinants in equation (45) can be written in a very simple way in the thermodynamic limit:

$$\frac{\det \Psi'(\{\lambda\}, \{\xi\})}{\det \Phi'(\{\lambda\})} = \frac{1}{M^m} \prod_{a=1}^m \tilde{\rho}_{tot}^{-1}(\lambda_{N-m+a}) \det S(\{\lambda_{N-m+1}, \dots, \lambda_N\}, \{\xi\}) + O(M^{-\infty}) \quad (52)$$

where

$$S_{ab} = \tilde{\rho} \left(\lambda_{N-m+a} - \xi_b + \frac{\eta}{2} \right) \quad (53)$$

Using the relations above, we finally obtain the following representation for the emptiness formation probability:

$$\begin{aligned} \tau(m) &= \frac{1}{M^m \prod_{k < l} \sinh(\xi_k - \xi_l)} \sum_{a_m=1}^N \sum_{a_{m-1}=1}^N \dots \sum_{a_1=1}^N H(\{\lambda_{a_1}, \dots, \lambda_{a_m}\}, \{\xi\}) \\ &\times \prod_{j=1}^m \tilde{\rho}_{tot}^{-1}(\lambda_{a_j}) \end{aligned} \quad (54)$$

where the function H is

$$\begin{aligned} H(\{\lambda_{a_1}, \dots, \lambda_{a_m}\}, \{\xi\}) &= \frac{1}{\prod_{k > l} \sinh(\lambda_{a_k} - \lambda_{a_l} + \eta)} \det S(\{\lambda_{a_1}, \dots, \lambda_{a_m}\}, \{\xi\}) \\ &\times \prod_{j=1}^m \left(\prod_{k=1}^{j-1} \sinh(\lambda_{a_j} - \xi_k + \eta) \prod_{k=j+1}^m \sinh(\lambda_{a_j} - \xi_k) \right) + O(M^{-\infty}) \end{aligned} \quad (55)$$

Note that when two indices coincide ($a_j = a_k$ ($j \neq k$)) the determinant of the matrix S vanishes.

In the thermodynamic limit for the ground state, we have:

$$\tau(m) = \frac{1}{\prod_{k < l} \sinh(\xi_k - \xi_l)} \int_{-\tilde{\Lambda}}^{\tilde{\Lambda}} d\lambda_1 \dots \int_{-\tilde{\Lambda}}^{\tilde{\Lambda}} d\lambda_m H(\{\lambda_1, \dots, \lambda_m\}, \{\xi\}) \quad (56)$$

where $\tilde{\Lambda} = \begin{cases} \Lambda & \text{for } -1 < \Delta \leq 1 \\ -i\Lambda & \text{for } \Delta > 1 \end{cases}$.

In the massless case the determinant of matrix S can be expressed as a Cauchy determinant,

$$\det S = \left(\frac{i}{2\zeta} \right)^m \frac{\prod_{k < l} \sinh \frac{\pi}{\zeta} (\xi_k - \xi_l) \prod_{a > b} \sinh \frac{\pi}{\zeta} (\lambda_a - \lambda_b)}{\prod_{a=1}^m \prod_{k=1}^m \sinh \frac{\pi}{\zeta} (\lambda_a - \xi_k)} \quad (57)$$

The emptiness formation probability has in this case the following form:

$$\begin{aligned} \tau(m) &= \prod_{k < l} \frac{\sinh \frac{\pi}{\zeta} (\xi_k - \xi_l)}{\sinh(\xi_k - \xi_l)} \int_{-\infty}^{\infty} i \frac{d\lambda_1}{2\zeta} \dots \int_{-\infty}^{\infty} i \frac{d\lambda_m}{2\zeta} \prod_{a > b} \frac{\sinh \frac{\pi}{\zeta} (\lambda_a - \lambda_b)}{\sinh(\lambda_a - \lambda_b - i\zeta)} \\ &\times \prod_{a=1}^m \prod_{k=1}^m \frac{1}{\sinh \frac{\pi}{\zeta} (\lambda_a - \xi_k)} \\ &\times \prod_{j=1}^m \left(\prod_{k=1}^{j-1} \sinh(\lambda_j - \xi_k - i\zeta) \prod_{k=j+1}^m \sinh(\lambda_j - \xi_k) \right) \end{aligned} \quad (58)$$

In the homogeneous case ($\xi_j = -i\zeta/2, \forall j$) we have for the emptiness formation probability:

$$\begin{aligned} \tau(m) = & (-1)^m \left(-\frac{\pi}{\zeta}\right)^{m(m+1)/2} \int_{-\infty}^{\infty} \frac{d\lambda_1}{2\pi} \cdots \int_{-\infty}^{\infty} \frac{d\lambda_m}{2\pi} \prod_{a>b} \frac{\sinh \frac{\pi}{\zeta}(\lambda_a - \lambda_b)}{\sinh(\lambda_a - \lambda_b - i\zeta)} \\ & \times \prod_{j=1}^m \frac{\sinh^{j-1} \left(\lambda_j - i\frac{\zeta}{2}\right) \sinh^{m-j} \left(\lambda_j + i\frac{\zeta}{2}\right)}{\cosh^m \frac{\pi}{\zeta} \lambda_j} \end{aligned} \quad (59)$$

In the massless case, the determinant of matrix S is more complicated but can still be expressed in terms of the Theta functions:

$$\det S = g_m \left(-\frac{1}{2\pi}\right)^m \frac{\prod_{j<k} \theta_1(i\lambda_j - i\lambda_k) \theta_1(i\xi_k - i\xi_j)}{\prod_{j,k=1}^m \theta_1(i\lambda_j - i\xi_k)} \theta_2 \left(\sum_{j=1}^m (i\lambda_j - i\xi_j)\right) \quad (60)$$

where

$$g_m = \prod_{n=1}^{\infty} \left(\frac{1-q^{2n}}{1+q^{2n}}\right)^2 \left(2q^{\frac{1}{4}} \prod_{n=1}^{\infty} (1-q^{2n})^3\right)^{m-1} \quad (61)$$

We change the variables : $\beta = i\xi, \zeta = -\eta$ thus for the emptiness formation probability:

$$\begin{aligned} \tau(m) = & g_m \prod_{k<l} \frac{\theta_1(\beta_k - \beta_l)}{\sin(\beta_k - \beta_l)} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} i \frac{d\lambda_1}{2\pi} \cdots \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} i \frac{d\lambda_m}{2\pi} \theta_2 \left(\sum_{j=1}^m (\lambda_j - \beta_j)\right) \\ & \times \prod_{a>b} \frac{\theta_1(\lambda_a - \lambda_b)}{\sin(\lambda_a - \lambda_b - i\zeta)} \prod_{a=1}^m \left(\prod_{k=1}^m \frac{1}{\theta_1(\lambda_a - \beta_k)} \prod_{k=1}^{a-1} \sin(\lambda_a - \beta_k - i\zeta) \prod_{k=a+1}^m \sin(\lambda_a - \beta_k)\right) \end{aligned} \quad (62)$$

In the homogeneous limit $\beta_j = -i\frac{\zeta}{2}$, we have:

$$\begin{aligned} \tau(m) = & \prod_{n=1}^{\infty} \left(\frac{1-q^{2n}}{1+q^{2n}}\right)^2 \left(2q^{\frac{1}{4}} \prod_{n=1}^{\infty} (1-q^{2n})^3\right)^{\frac{m(m+1)}{2}-1} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} i \frac{d\lambda_1}{2\pi} \cdots \\ & \times \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} i \frac{d\lambda_m}{2\pi} \theta_2 \left(\sum_{j=1}^m \left(\lambda_j + i\frac{\zeta}{2}\right)\right) \prod_{a>b} \frac{\theta_1(\lambda_a - \lambda_b)}{\sin(\lambda_a - \lambda_b - i\zeta)} \\ & \times \prod_{j=1}^m \frac{\sin^{j-1} \left(\lambda_j - i\frac{\zeta}{2}\right) \sin^{m-j} \left(\lambda_j + i\frac{\zeta}{2}\right)}{\theta_1^m \left(\lambda_j + i\frac{\zeta}{2}\right)} \end{aligned} \quad (63)$$

$$\int_{-\infty}^{\infty} \frac{d\lambda_1}{2\pi} \int_{-\infty}^{\infty} \frac{d\lambda_2}{2\pi} \frac{\sinh \frac{\pi}{\zeta}(\lambda_2 - \lambda_1)}{\sinh(\lambda_2 - \lambda_1 - i\zeta)} \cdot \frac{\sinh \left(\lambda_1 + i\frac{\zeta}{2}\right)}{\cosh^2 \frac{\pi}{\zeta} \lambda_1} \cdot \frac{\sinh \left(\lambda_2 - i\frac{\zeta}{2}\right)}{\cosh^2 \frac{\pi}{\zeta} \lambda_2} \quad (64)$$

4 Correlation function

To calculate an arbitrary m-point correlation function:

$$F_m(\{\epsilon_j, \epsilon'_j\}) = \frac{\langle \psi_g | \prod_{j=1}^m E_j^{\epsilon'_j, \epsilon_j} | \psi_g \rangle}{\langle \psi_g | \psi_g \rangle} \quad (65)$$

By quantum inverse scattering method, we can represent the local operators in terms of monodromy matrices:

$$E_j^{\epsilon'_j, \epsilon_j} = \prod_{k=1}^{j-1} (A(\xi_k) + D(\xi_k)) T_{\epsilon_j, \epsilon'_j}(\xi_j) \prod_{k=j+1}^M (A(\xi_k) + D(\xi_k)) \quad (66)$$

Thus we get :

$$F_m(\{\epsilon_j, \epsilon'_j\}) = \phi_m(\{\lambda\}) \frac{\langle \psi_g | T_{\epsilon_1, \epsilon'_1}(\xi_1) \dots T_{\epsilon_m, \epsilon'_m}(\xi_m) | \psi_g \rangle}{\langle \psi_g | \psi_g \rangle} \quad (67)$$

where $\phi_m(\{\lambda\})$ is given by

$$\phi_m(\{\lambda\}) = \prod_{j=1}^m \prod_{a=1}^N \frac{\sinh(\lambda_a - \xi_j)}{\sinh(\lambda_a - \xi_j + \eta)} \quad (68)$$

Some indices notations are explained in the primary paper.

Consider the ordered product acting on the ground Bethe state:

$$\langle 0 | \prod_{k=1}^N C(\lambda_k) T_{\epsilon_1, \epsilon'_1}(\lambda_{N+1}) \dots T_{\epsilon_m, \epsilon'_m}(\lambda_{N+m}) = \sum_{\{a_j, a'_j\}} G_{\{a_j, a'_j\}}(\lambda_1, \dots, \lambda_{N+m}) \langle 0 | \prod_{b \in A_{m+1}} C(\lambda_b) \quad (69)$$

The summation is taken over the indices a_j for $j \in \alpha^-$ and a'_j for $j \in \alpha^+$ such that

$$1 \leq a_j \leq N + j, \quad a_j \in A_j, \quad 1 \leq a'_j \leq N + j, \quad a'_j \in A'_j \quad (70)$$

The functions $G_{\{a_j, a'_j\}}(\lambda_1, \dots, \lambda_{N+m})$ is given by:

$$\begin{aligned} G_{\{a_j, a'_j\}}(\lambda_1, \dots, \lambda_{N+m}) &= \prod_{j \in \alpha^-} d(\lambda_{a_j}) \frac{\prod_{\substack{b=1 \\ b \in A_j}}^{N+j-1} \sinh(\lambda_{a_j} - \lambda_b + \eta)}{\prod_{\substack{b=1 \\ b \in A'_j}}^{N+j} \sinh(\lambda_{a_j} - \lambda_b)} \\ &\times \prod_{j \in \alpha^+} a(\lambda_{a'_j}) \frac{\prod_{\substack{b=1 \\ b \in A'_j}}^{N+j-1} \sinh(\lambda_b - \lambda_{a'_j} + \eta)}{\prod_{\substack{b=1 \\ b \in A_{j+1}}}^{N+j} \sinh(\lambda_b - \lambda_{a'_j})} \end{aligned} \quad (71)$$

We can then consider the scalar products:

$$\frac{\langle 0 | \prod_{b \in A_{m+1}} C(\lambda_b) \prod_{k=1}^N B(\lambda_k) | 0 \rangle}{\langle 0 | \prod_{k=1}^N C(\lambda_k) \prod_{k=1}^N B(\lambda_k) | 0 \rangle} \quad (72)$$

Finally we obtain:

$$F_m(\{\epsilon_j, \epsilon'_j\}) = \frac{1}{\prod_{k < l} \sinh(\xi_k - \xi_l)} \sum_{\{a_j, a'_j\}} H_{\{a_j, a'_j\}}(\lambda_1, \dots, \lambda_{N+m}) \quad (73)$$

The functions H can be obtained using scalar products formulas. For convenience, we introduce the following set of indices:

$$\{b_1, \dots, b_m\} = \{a'_{j_{max}}, \dots, a'_{j_{min}}, a_{j_{min}}, \dots, a_{j_{max}}\} \quad (74)$$

Note that for the XXZ model $a(\lambda) = 1$ and $d(\xi_k) = 0$. Thus we obtain $a_j \leq N$, $\forall j \in \alpha^-$, otherwise $d(\lambda_{a_j}) = 0$. We give the formula of H below.

$$\begin{aligned} H_{\{a_j, a'_j\}}(\{\lambda\}) &= \frac{(-1)^{s'}}{\prod_{k > l} \sinh(\lambda_{b_k} - \lambda_{b_l} + \eta)} \\ &\times \prod_{j \in \alpha^+} \left(\prod_{k=1}^{j-1} \sinh(\lambda_{a'_j} - \xi_k - \eta) \prod_{k=j+1}^m \sinh(\lambda_{a'_j} - \xi_k) \right) \\ &\times \prod_{j \in \alpha^-} \left(\prod_{k=1}^{j-1} \sinh(\lambda_{a_j} - \xi_k + \eta) \prod_{k=j+1}^m \sinh(\lambda_{a_j} - \xi_k) \right) \\ &\times (\det M(\{b_k\}) + O(M^{-\infty})) \end{aligned} \quad (75)$$

where

$$\begin{aligned} b_l > N, \quad M_{lk} &= -\delta_{b_l - N, k} \\ b_l \leq N, \quad M_{lk} &= \frac{\tilde{\rho}(\lambda_{b_l} - \xi_k + \frac{\eta}{2})}{\tilde{\rho}_{tot}(\lambda_{b_l})} \end{aligned} \quad (76)$$

Note that $H = 0$ if $b_j = b_k$, $j \neq k$ or if $a'_j > N + j$ or if $a_j > N$ ($d = 0$). We can rewrite F as

$$F_m(\{\epsilon_j, \epsilon'_j\}) = \frac{1}{\prod_{k < l} \sinh(\xi_k - \xi_l)} \sum_{b_1=1}^{N+m} \dots \sum_{b_{s'}=1}^{N+m} \sum_{b_{s'+1}=1}^N \dots \sum_{b_m=1}^N H_{\{a_j, a'_j\}}(\{\lambda\}) \quad (77)$$

The sum over $1 \leq b_j \leq N$ can be replaced by integrals over rapidities in the ground state.